Sibasish Barik

Indore, Madhya Pradesh

+91-9556597254 | sibasishbarik05@gmail.com | LinkedIn | GitHub | Portfolio

Education

Indian Institute of Technology Indore Aug. 2023 - May 2027

Bachelor of Technology

Profile Summary

Robotics and UAV enthusiast who loves turning ideas into machines that actually fly, drive, and make decisions on their own. I am a Civil Engineering student who fell for robotics and mechanical design, and I am excited to join a team where I can keep building, testing, and learning from real-world systems.

Experience

•Center for System Design, NITK Surathkal *May 2026 - Jul. 2026*

Research Intern NITK Surathkal | Certificate

- Developed the URDF and a complete autonomous flight simulation framework for a Cessna fixed-wing aircraft using ROS 2, Gazebo Harmonic, and ArduPilot SITL integrated via MAVLink and MAVROS.
- Deployed advanced waypoint mission management and a custom reactive obstacle avoidance controller, inspired by a hybrid of Artificial Potential Fields (APF) and Vector Field Histogram (VFH) methods, to dynamically navigate complex obstacle fields under fixed-wing kinematic constraints.
- Maximised low-altitude flight reliability through custom physics-engine collision boundary constraints, achieving stable ultra-low altitude simulation and automated trajectory recovery.

•Charak Center for Digital Healthcare *May 2025 - Jul. 2025*

SIDDHI 3.0 Research Intern IIT Indore | Certificate

- Engineered and field-tested an autonomous medical delivery UAV prototype integrating ArUco-based computer vision for sub-meter precision landing and thermal cold-box payload architectures for temperature-sensitive supplies, aligning system design with operational constraints identified through workflow assessments at regional healthcare facilities for a 20km operation range.

Projects

•UAV-Based Water Quality Monitoring and Sampling System for Rapid Environmental Assessment *Present*

Dr. Mayur Shirish Jain, Environmental Engineering Lab

- Conducting literature review and proof-of-concept development for an autonomous UAV-based water quality monitoring and sampling system.
- Designing a tether-assisted sensing mechanism integrating pH, EC, TDS, temperature, and turbidity sensors for geo-referenced, depth-specific measurements.
- Evaluating commercial viability through Techno-Economic Analysis using CAPEX/OPEX, NPV, IRR, Benefit-Cost Ratio, Payback Period, and sensitivity analysis.

•Autonomous Dual-UAV System for Disaster Management *Apr. 2025 - Jan. 2026*

NIDAR Challenge 2025, MeitY Github | Documentation

- Led a 10-member team to design, build and fly a dual-drone UAV system using Pixhawk 6C, Cube Orange that autonomously surveys a 30-ha zone and delivers 10 X 200g payloads.
- Geotagged survivors by fusing F9P GNSS with onboard YOLOv8s/BoT-SORT detection on a Jetson Orin Nano, cutting duplicate and false tags via spatial filtering.
- Built the autonomy stack: KML-based lawnmower coverage planning, capacity-batched TSP routing, py_trees behavior-tree mission logic, and Hailo-8 visual servoing for 100cm payload-drop accuracy.

•Docking and Charging System for Autonomous UAV *Aug. 2025 - Present*

IRoC-U 2026, ISRO Documentation

- Led end-to-end CAD design, prototyping, and fabrication of a custom UAV docking station integrating mechanical alignment and contact-based autonomous charging.
- Developed a self-aligning docking mechanism using synchronised conductive sliders to correct residual landing misalignment.
- Integrated ArUco-based visual alignment and Raspberry Pi-controlled charging hardware, demonstrating the complete docking-to-charging sequence and supported UAV hardware assembly and electronics integration.

•Simulation and Optimisation of Autonomous Mobile Robot Navigation using Nav2 *Aug. 2026 - Sep. 2026*

Self Project GitHub

- Simulated an autonomous differential-drive mobile robot using **ROS 2 Jazzy, Gazebo, and Nav2** for map-based navigation and obstacle avoidance.
- Configured **Adaptive Monte-Carlo localisation**, LiDAR-based perception, odometry, and coordinate-frame transformations for robot pose estimation.
- Integrated **NavFn global planning, layered costmaps, and the MPPI local controller** to generate and execute obstacle-aware navigation commands.
- Tuned controller parameters, prediction horizons, velocity constraints, and trajectory critics; debugged DWB control failures and achieved successful goal navigation using MPPI.

•Autonomous PID-Controlled Light Detecting and Following Drone *Apr. 2025 - Nov. 2025*

Dr. Sumit Gautam, Dr. Devesh Mishra (SwaYaan Lab, IIT Indore) Github | Documentation

- Developed an autonomous line-following UAV system that uses real-time computer vision on a Raspberry Pi to track an LED-marked path with an ArduPilot flight controller.
- Implemented dual PID controllers (lateral position and yaw rate) with anti-windup, derivative filtering, and output saturation, replacing discrete step commands with continuous MAVLink velocity control for smoother flight.

•Control System for Drone Stabilisation and Precision Motion *Aug. 2025 - Present*

Inter IIT Techmeet 14.0, Problem Statement by Drona Aviation Documentation

- Designed and implemented a cascaded PID control architecture for drone stabilisation and precision motion.
- Developed sensor-fusion and filtering algorithms using a Linear Kalman Filter and BiQuad filtering to reduce IMU and altitude-sensor noise, enabling smoother velocity estimation and stable flight behavior.
- Secured 8th position in the Low-Prep Problem Statement among 23 IITs.

•OpenMower - Autonomous Lawnmower with Integrated Robotic Arm *Aug. 2025 - Present*

Dr. Vimal Bhatia

- Integrated robotic sensors and hardware-software interfaces using Arduino and Raspberry Pi for an autonomous lawnmower platform.
- Developing a ROS 2 navigation node to process NMEA GPS data and support GPS-guided outdoor navigation.
- Containerising the robotics codebase with Docker for reproducible deployment and execution on Raspberry Pi.

•Autonomous Electric Vehicle Powertrain Electronics *Aug. 2024 - Apr. 2025*

Center for Electric Vehicle and Intelligent Transport Systems (CEVITS)

- Designed and built a 72V 100Ah Li-Phosphate battery pack pre-charge circuit, using industry-grade relays for an electric vehicle powertrain.
- Integrated a industry-grade Kelly KLS motor controller with a PMSM traction motor to enable electric vehicle motor operation.

Technical Skills

- **Programming Languages**: Python (OOP), C/C++, Arduino programming, JavaScript*, MATLAB*
- **Hardware**: Raspberry Pi, NVIDIA Jetson, Arduino, ESP32, Teensy, HW/SW integration, sensor fusion, Pixhawk, ArduPilot
- **Software Tools & Frameworks**: ROS 2, Nav2, SLAM, OpenCV, Gazebo, ArduPilot, PX4-Autopilot, DroneKit-Python, MAVSDK-Python, PyMAVLink, MS Excel, Docker, Git
- **CAD & Analysis**: Autodesk Fusion 360 & AutoCAD, LaserCAD, LightBurn, DS SolidWorks, Ansys WB
- **Operating Systems**: Windows, Linux*

** Elementary proficiency*

Key courses taken Certificates

- **Mathematics**: Linear Algebra, Calculus, Differential Equations, Probability and Statistics, Numerical Methods
- **Civil and Mechanical**: Strength of Materials, Fluid Mechanics, Engineering Mechanics, Structural Analysis, Introduction to FEM, Computational Fluids and Structures, Flight Mechanics and Classical Control, Introduction to Space Engineering, Introduction to Astronomy
- **Additional Courses**: SOLIDWORKS Associate Course, SOLIDWORKS Professional Course, Aircraft Systems and Avionics, Aircraft Fundamentals, MATLAB Onramp, AI/ML

Positions of Responsibility

- **Head**, Aeromodelling Club, IIT Indore — *Apr. 2025 - Apr. 2026*
- **Inventory Head**, Tinkerers' Lab, IIT Indore — *Apr. 2025 - Apr. 2026*
- **Team Lead**, NIDAR 2025 — *2025-26 Edition*
- **Design Lead**, ISRO Robotics Challenge - URSC 2026 — *2026 Edition*
- **Team Lead**, NXP AIM 2024 — *2024 Edition*
- **Member**, Intelligent Vehicle Design and Control Club, IIT Indore — *Apr. 2024 - Present*
- **Member**, Astronomy Club, IIT Indore — *Nov. 2023 - Present*
- **Member**, Animal Welfare Team, IIT Indore — *Sept. 2023 - Present*

Achievements

Certificates

- **3rd Rank Overall**, Inter IIT Techmeet 14.0 — *2026*
- **32nd Rank**, NIDAR Challenge 2025 — *2025*
- **Top 74 Teams**, IRoC-U 2026 — *2026*
- **Gold Medal**, IITI Summer of Code, Concreate Domain — *2024*
- **Silver Medal**, Inter IIT Civil Conclave — *2024*